US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250282160
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	WADA; ITARU et al.

RECORDING DEVICE

Abstract

A recording device includes: a first fixing member; a first contact portion movable between a first pressing position to press a first shaft hole of a first roll sheet radially outward and a first non-pressing position located on a radially inward side of the first pressing position; a cam moving the first contact portion between the first pressing position and the first non-pressing position; a second fixing member attachable to or detachable from the first fixing member; and a second contact portion movable between a second pressing position to press a second shaft hole of a second roll sheet radially outward and a second non-pressing position located on a radially inward side of the second pressing position. In a state where the second fixing member is attached to the first fixing member, the cam moves the second contact portion between the second pressing position and the second non-pressing position.

Inventors: **WADA; ITARU (Kanagawa, JP), NANJI; YUYA (Kanagawa, JP), MURATA; RYOSUKE (Tokyo, JP)**

Applicant: **CANON KABUSHIKI KAISHA** (Tokyo, JP)

Family ID: **96948402**

Appl. No.: **19/060601**

Filed: **February 21, 2025**

Foreign Application Priority Data

JP	2024-033235	Mar. 05, 2024

Publication Classification

Int. Cl.: **B41J25/308** (20060101); **B41J15/02** (20060101)

U.S. Cl.:

CPC **B41J25/3082** (20130101); **B41J15/02** (20130101);

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a recording device which performs recording on a sheet supplied from a roll sheet held by a holding device.

Description of the Related Art

[0002] There is a recording device, which includes a holding device holding a roll sheet, which is a sheet rolled around a hollow shaft having a shaft hole, and performs recording on the sheet supplied from the roll sheet held by the holding device. According to a recording device disclosed in Japanese Patent No. 3928705, a roll sheet pressor is attached to the shaft of the roll sheet, the roll sheet held by the roll sheet pressor is attached to a spindle, and the spindle is supported by spindle bearings of the recording device main unit, whereby the roll sheet is set in the recording device. The inner diameter of the shaft hole of the shaft of the roll sheet may be different depending on the roll sheet.

[0003] The roll sheet pressor of the recording device according to Japanese Patent No. 3928705 includes a flange of which shaft core corresponding to a small diameter shaft hole protrudes, and an adapter corresponding to a large diameter shaft hole is attached to the shaft core, so as to support roll sheets having different inner diameters of the shaft holes. The shaft core and the adapter have a leaf spring which radially bends, and when the shaft hole of the roll sheet is inserted into the shaft core and the adapter, the shaft of the roll sheet is firmly held by the elastic force generated by the bending of the leaf spring.

[0004] In the case of the recording device according to Japanese Patent No. 3928705 however, the shaft core of the roll sheet pressor needs to be attached to/detached from the shaft hole of the roll sheet while bending the leaf spring, hence a force for bending the leaf spring against the elastic force of the leaf spring is required when the roll sheet pressor is attached/detached. Therefore, operability of the attachment/detachment of the roll sheet pressor need to be improved.

SUMMARY OF THE INVENTION

[0005] In the recording device having the holding device which can support roll sheets of which inner diameters of the respective shaft holes are different, the present invention improves operability of the attachment/detachment of the roll sheet to/from the holding device.

[0006] The present invention is a recording device, comprising: [0007] a holding device configured to hold a roll sheet that is a sheet rolled around a shaft having a shaft hole; and [0008] a recording unit configured to perform recording on the sheet supplied from the holding device that holds the roll sheet, wherein [0009] the holding device further comprising: [0010] a first fixing member including a first core configured to be inserted into a first shaft hole of a first roll sheet, the first shaft hole having a first inner diameter; [0011] a first contact portion formed of an elastic body disposed in the first fixing member, the first contact portion being configured to move between a first pressing position where the first contact portion contacts an inner peripheral surface of the first shaft hole and presses the inner peripheral surface radially outward and a first non-pressing position located on a radially inward side of the first pressing position; [0012] a cam configured to move the first contact portion between the first pressing position and the first non-pressing position; [0013] a second fixing member configured to be attached to or detached from the first fixing member, the second fixing member including a second core configured to be inserted into a second shaft hole of a second roll sheet, the second shaft hole having a second inner diameter larger than the first inner diameter; and [0014] a second contact portion formed of an elastic body disposed in the second

fixing member, the second contact portion being configured to move between a second pressing position where the second contact portion contacts an inner peripheral surface of the second shaft hole and presses the second shaft hole radially outward and a second non-pressing position located on a radially inward side of the second pressing position, wherein [0015] in a state where the second fixing member is attached to the first fixing member, the cam moves the second contact portion between the second pressing position and the second non-pressing position.
[0016] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0017] FIG. **1** is a perspective view of a recording device of an embodiment;
[0018] FIG. **2** is a perspective view of a recording device at the time of performing roll set;
[0019] FIG. **3** is a schematic cross-sectional view of a conveying path of s sheet in the recording device;
[0020] FIGS. **4**A to **4**C are diagrams for describing a roll support member in the recording device;
[0021] FIG. **5** is an external view of a spindle flange of Embodiment 1;
[0022] FIG. **6** is an external view of a first contact portion of Embodiment 1;
[0023] FIGS. **7**A and **7**B are external views of a cam of Embodiment 1;
[0024] FIG. **8**A is a schematic diagram of a first position when a 2-inch roll is attached according to Embodiment 1;
[0025] FIG. **8**B is a schematic diagram of a first position when a 2-inch roll is attached according to Embodiment 1;
[0026] FIG. **9**A is a schematic diagram of a second position when a 2-inch roll is attached according to Embodiment 1;
[0027] FIG. **9**B is a schematic diagram of a second position when a 2-inch roll is attached according to Embodiment 1;
[0028] FIG. **10** is an external view of an attachment of Embodiment 1;
[0029] FIG. **11** is an external view when the attachment of Embodiment 1 is attached to the spindle flange;
[0030] FIG. **12** is an external view of a second contact portion of Embodiment 1;
[0031] FIG. **13**A is a schematic diagram of a first position when a 3-inch roll is attached according to Embodiment 1;
[0032] FIG. **13**B is a schematic diagram of a first position when a 3-inch roll is attached according to Embodiment 1;
[0033] FIG. **14**A is a schematic diagram of a second position when a 3-inch roll is attached according to Embodiment 1;
[0034] FIG. **14**B is a schematic diagram of a second position when a 3-inch roll is attached according to Embodiment 1;
[0035] FIG. **15** is an external view of a spindle flange of Embodiment 2;
[0036] FIG. **16** is an external view of a first contact portion of Embodiment 2;
[0037] FIG. **17** is an external view of a cam of Embodiment 2;
[0038] FIG. **18** is an external view of a grip portion of Embodiment 2;
[0039] FIG. **19** is a schematic diagram of a first position when a 2-inch roll is attached according to Embodiment 2;
[0040] FIG. **20** is a schematic diagram of a second position when a 2-inch roll is attached according to Embodiment 2;
[0041] FIG. **21** is an external view of an attachment of Embodiment 2;

[0042] FIG. **22** is an external view when the attachment of Embodiment 2 is attached to the spindle flange;
[0043] FIG. **23** is an external view of a second contact portion of Embodiment 2;
[0044] FIG. **24** is a schematic diagram of a first position when a 3-inch roll is attached according to Embodiment 2;
[0045] FIG. **25** is a schematic diagram of a second position when a 3-inch roll is attached according to Embodiment 2;
[0046] FIG. **26** is an external view of a non-reference side spindle flange of Embodiment 3;
[0047] FIG. **27** is an external view of a spindle lock flange of Embodiment 3;
[0048] FIG. **28** is an external view of a spindle lock lever of Embodiment 3;
[0049] FIG. **29** is an external view when the spindle lock is unlocked according to Embodiment 3;
[0050] FIG. **30** is an external view when the spindle lock is locked according to Embodiment 3;
[0051] FIG. **31** is an external view of a non-reference side spindle flange when an intermediate link mechanism is unlocked according to Embodiment 3; and
[0052] FIG. **32** is an external view of a non-reference side spindle flange when the intermediate link mechanism is locked according to Embodiment 3.

DESCRIPTION OF THE EMBODIMENTS

[0053] Embodiments of the present invention will now be described with reference to the drawings. Dimensions, materials and shapes of components described in the following embodiments and relative positions thereof should be appropriately changed in accordance with the configuration of the device to which the present invention is in applied, and with various conditions, and are not intended to limit the scope of the present invention to the following embodiments.

Basic Configuration of Device

[0054] FIGS. **1** to **3** are diagrams for describing a basic configuration of an embodiment. A recording device **100** of the embodiment includes a roll set portion **200**, which is a holding device to hold a roll sheet (rolled sheet), which is a sheet rolled around a hollow shaft having a shaft hole. The recording device **100** is an inkjet recording device which records an image on a sheet type recording medium (sheet) supplied from the roll set portion **200**. In this embodiment, an example of applying the present invention to the inkjet recording device will be described, but the present invention is applicable to any recording device having the roll set portion **200**, and the printing system thereof is not limited to an inkjet recording device. In the following description, it is assumed that the X direction is a direction parallel with the axial direction (shaft line direction) of the hollowed shaft of the roll sheet, the Y direction is a direction that is vertical to the axial direction and parallel with the horizontal direction, and the Z direction is a vertical direction.

[0055] FIG. **1** is a perspective view of the recording device **100** which can record an image on a sheet supplied from a roll sheet generated by rolling up a sheet **1**, and can wind up the recorded sheet **1**. FIG. **2** is a perspective view of the recording device **100** at the time of performing roll set in the recording device **100**. By setting a sheet delivery guide portion **500** to an open state, a roll sheet can be set in a sheet feeding portion. The recording device **100** includes the roll set portion **200**, which is a holding device to hold the roll sheet.

[0056] When the sheet delivery guide portion **500** is in open state, a roll sheet can be set in the roll set portion **200**. When the sheet delivery guide portion **500** is in closed state, the sheet drawn out of the roll sheet that is set in the roll set portion **200** reaches a print portion via a sheet conveying portion, and after an image is printed on the sheet by the print portion, the sheet is discharged to the sheet delivery guide portion **500**. The user can input various commands and the like to the recording device **100**, to specify the size of the roll sheet, the type of the roll, and the like, using various switches provided on the operation panel **28**.

[0057] FIG. **3** is a schematic cross-sectional view of a key portion of the recording device **100**. The sheet **1** drawn out from a roll sheet R, which is set in the roll set portion **200**, is connected to a winding portion **600** and wound up. The sheet **1** drawn out from the roll sheet R, which is set in the roll set portion **200**, is conveyed to a print portion **400** (a recording portion that can print the image) via a sheet conveying portion **300**. The print portion **400** prints an image on the sheet **1** by ejecting ink from an inkjet type print head **8**. The print head **8** ejects ink from an ejection port using an ejection energy generation element, such as an electro-thermal conversion element (heater) and a piezoelectric element. In the case of using an electro-thermal conversion element, ink is foamed by the heating thereof, and ink can be ejected from the ejection port using the foaming energy thereof. The print head **8** is not limited to the inkjet type, and the print type of the print portion **400** is also not limited, and may be a serial scan type or a full-line type, for example. In the case of the serial scan type, an image is printed using the conveying operation of the sheet **1**, and scanning with the print head **8** in a direction intersecting with the conveying direction of the sheet **1**. In the case of the full-line type, an image is printed while conveying the sheet **1** continuously using a long print head **8**, which extends in a direction intersecting with the conveying direction of the sheet **1**.

[0058] A shaft-shaped roll support member **2** is inserted into a hollow shaft **40** (also called a "sheet tube") of the roll sheet R, and the roll support member **2** is driven by a roll driving motor to perform normal rotation or reverse rotation. Thereby the roll sheet R performs normal rotation or reverse rotation (arrow C**1** or C**2** direction) while the center portion thereof is held. The roll set portion **200** includes: a driving portion **3**, an arm member (moving body) **4**, an arm rotation shaft **5**, a first sheet sensor **6**, an oscillation member **7**, a driven rotator (pressure contact body) **8**, a separation flapper (upper guide body) **9**, and a flapper rotation shaft **10**.

[0059] A sheet feeding side conveying guide **11** supports the front and rear surfaces of the sheet **1** drawn from the roll set portion **200**, and guides this sheet **1** to the print portion **400** in this state. A conveying roller **12** performs normal rotation or reverse rotation in the arrow D**1** or D**2** direction by the conveying roller driving motor. A nip roller **13** can be rotate-driven by the rotation of the conveying roller **12**, and can be separated from the conveying roller **12** and the nip force can be adjusted by a nip roller specification motor (not illustrated). The conveying roller **12** is rotated when a second sheet sensor **14** detects the front end of the sheet **1**. The speed of conveying the sheet **1** by the conveying roller **12** is set to be faster than the speed of drawing the sheet **1** by rotation of the roll sheet R, whereby back tension is applied to the sheet **1** so that the sheet **1** is conveyed in a stretched state. As a result, sagging of the sheet **1** is prevented, and the generation of creases of the sheet **1** and the generation of conveying errors can be suppressed.

[0060] A platen **15** of the print portion **400** adsorbs the rear face of the sheet **1** via suction holes formed on the platen **15** using negative pressure generated by a suction fan **16**. Thereby the position of the sheet **1** can be controlled along the surface of the platen **15**, and the print head **8** can print the image at high precision. A cutter **17** cuts the sheet **1** on which the image is printed.

[0061] A sheet delivery side conveying guide **18** installed in the sheet delivery guide portion **500** supports the rear face of the sheet **1** drawn from the print portion **400**, and guides the sheet **1** to the winding portion **600** in this state. Here the front end of the sheet **1** is fixed at the hollow shaft **40** which is set in the winding portion **600**, and the hollow shaft **40** that is set is rotated in accordance with the conveying speed of the conveying roller **12**, whereby the sheet **1** which was printed by the print portion **400** can be wound up continuously. A driven roller **33** is installed at the edge of the winding side of the sheet delivery side conveying guide **18**. Thereby damage to the medium that may be caused by bending at the driven roller **33** can be prevented, and conveying resistance at a bent portion can be decreased, which prevents the generation of major deflection between the conveying roller **12** and the driven roller **33**. Because of this configuration, where the sheet **1** is conveyed along the sheet delivery side conveying guide **18**, the sheet delivery side conveying guide **18** can be heated using a heater (not illustrated) when necessary, so as to assist thermally fixing the ink, ejected by the print portion **400**, to the sheet **1**. The sheet delivery side conveying guide **18** and the driven roller **33** can rotate around the sheet delivery guide rotation shaft **20**. Thereby when the roll sheet R is attached to/detached from the roll set portion **200**, the sheet delivery side conveying guide **18** and the driven roller **33** can be retracted upward by rotation, and the roll sheet R can be set without interfering with the sheet delivery guide portion **500**.

[0062] The main unit leg portions **27** support the weight of the sheet conveying portion **300** and the print portion **400** on the upper part of the main unit from below, and also support the weight of the roll sheet R via the roll set portion **200**. The main unit leg portions **27** contact the floor surface via casters, so as to release the weight of the recording device **100** to the floor surface. Linking portions **29** connect the main unit leg portions **27** on both sides in the sheet width direction, and prevent the falling of the main unit leg portions **27** and the like due to weight.

[0063] FIGS. **4**A, **4**B and **4**C are diagrams for describing a procedure to set the roll sheet R in the roll set portion **200** using the roll support member **2**. The roll support member **2** includes a spindle **19**, a first friction member **31**, a reference side spindle flange **21**, a non-reference side spindle flange **22**, and a spindle gear **23**. The reference side spindle flange **21** is disposed on one end of the spindle **19**, and the spindle gear **23** which rotates the spindle **19**, is disposed on the other end. The first friction member **31** is disposed on the inner side of the reference side spindle flange **21** and the non-reference side spindle flange **22**.

[0064] When the roll support member **2** is set on the roll sheet R, the non-reference side spindle flange **22** inserted into the spindle **19** is removed, then the spindle **19** is inserted into the hollow shaft **40** of the roll sheet R. The outer diameter of the spindle **19** is smaller than the inner diameter of the hollow shaft **40** of the roll sheet R, and a gap is created therebetween, hence the user can insert the spindle **19** into the hollow shaft **40** with very little force. When the edge of the roll sheet R on the right side in FIG. **4**A touches the reference side spindle flange **21**, the first friction member **31** on the inner side of the reference side spindle flange **21** is inserted into the hollow shaft **40** of the roll sheet R. Thereby the reference side spindle flange **21** is fixed to the roll sheet R. Then, with the spindle **19** being passed through the non-reference side spindle flange **22**, the first friction member **31** on the inner side of the non-reference side spindle flange **22** is inserted into the hollow shaft **40** of the roll sheet R. Thereby the non-reference side spindle flange **22** is fixed to the roll sheet R.

[0065] In this way, the roll sheet R is attached to the roll support member **2**, as illustrated in FIG. **4**B. Then, as illustrated in FIG. **4**C, both ends of the roll support member **2** are inserted into the spindle holders **24** of the roll set portion **200**, whereby the setting of the roll sheet R is completed.

[0066] It is assumed here that the outer diameters of the reference side spindle flange **21** and the non-reference side spindle flange **22** are about 170 mm, for example. The maximum outer diameter of the roll sheet R is about 180 mm, for example, and the inner diameter of the hollow shaft **40** is 2 inches (50.8 mm) or 3 inches (76.2 mm). The outer diameters of the reference side spindle flange **21** and the non-reference side spindle flange **22**, the maximum outer diameter of the roll sheet R, and the inner diameter of the hollow shaft **40** are not limited to the above-mentioned values.

[0067] The spindle holders **24** are disposed at positions corresponding to both ends of the spindle **19** respectively. The inner surface of the spindle holder **24** is formed in a U shape, and each end of the spindle **19** can be inserted from the opening side of the U shape. In a state where the roll support member **2** is inserted into the spindle holders **24**, the spindle gear **23** is connected to a roll driving motor via the driving gear **25** on the roll set portion **200** side. When the roll sheet R along with the roll support member **2** are driven by the roll driving motor in normal rotation or reverse rotation, the sheet **1** can be supplied. The spindle holders **24** play a role of feeding support portions which support the roll support member **2** so as to feed the roll sheet R, whereby the position of the roll sheet R, with respect to the recording device **100**, can be accurately determined. A roll sensor **26** detects the presence of the roll sheet R.

[0068] By using the spindle **19** to hold the roll sheet R like this, the roll sheet R is set in the roll set portion **200** by the spindle **19** and the spindle holders **24**, regardless the width of the sheet. Therefore, the roll sheet R having a width corresponding to various sizes less than the length of the spindle **19**, from such a large size as A**0** and A**1** to a small size, can be set. Further, the edge of the roll sheet R is inserted into the reference side spindle flange **21** fixed to the spindle **19**, hence the reference side edge position of the roll sheet R can be determined without deviation in the roll set portion **200**.

Embodiment 1

[0069] The procedure to attach the roll sheet R to the roll support member **2** was described with reference to FIGS. **4**A to **4**C, now the configuration of Embodiment 1, which allows to attach roll sheets R having different inner diameters of shift hole of the shafts to the roll set portion **200** (holding device), will be described with reference to FIGS. **5** to **14**A and **14**B.

[0070] First, attachment of a first roll sheet R**1** having a first shaft hole **41** of which inner diameter is 2 inches (first inner diameter) will be described.

[0071] FIG. **5** is an external view of a spindle flange. The reference side spindle flange **21** includes a flat reference side flange surface **131** and attachment installation holes **164**. The first friction member **31** is constituted of a reference side flange core **132** which protrudes from the reference side flange surface **131**, and 2 (a pair of) first contact portions **136** which are disposed symmetrically with respect to the shaft line of the spindle **19**.

[0072] FIG. **6** is an external view of the first contact portion **136**. The first contact portion **136** includes a fixing portion **150**, which is fixed to the reference side spindle flange **21**. The first contact portion **136** also includes a flat portion **152** to which a later mentioned first pressing portion **161** is pressed. The first contact portion **136** also includes a pair of interference portions **153**, which contact with the inner periphery of the first shaft hole **41** of the first roll sheet R**1**. Further, a rotating cam **160** is disposed inside the reference side spindle flange **21**.

[0073] FIGS. **7**A and **7**B are external views of the rotating cam **160**. The rotating cam **160** includes a lever portion **163**, the two first pressing portions **161** which are disposed symmetrically with respect to the shaft line of the spindle **19**, and 2 second pressing portions **162** which are also disposed symmetrically with respect to the shaft line of the spindle **19**. The rotating cam **160** has a cylindrical portion **169** which is coaxial with the spindle **19**, and the first pressing portions **161** and the second pressing portions **162** are disposed on the outer peripheral surface of the cylindrical portion **169**. The rotating cam **160** is rotatably supported inside the reference side spindle flange **21**, and can be switched between the attitude at the first position illustrated in FIG. **7**A and the attitude at the second position illustrated in FIG. **7**B.

[0074] FIG. **8**B is a cross-sectional view of the reference side spindle flange **21** sectioned at a plane including the shaft line of the first roll sheet R**1** (shaft line of the spindle **19**), and FIG. **8**A is a cross-sectional view at the A-A line in FIG. **8**B. FIGS. **8**A and **8**B indicate the attitude of the first contact portion **136** and the rotating cam **160** at the first position. In the attitude at the first position, the first pressing portion **161** and the flat portion **152** of the first contact portion **136** do not contact.

[0075] FIGS. **9**A and **9**B indicate the attitude of the first contact portion **136** and the rotating cam **160** at the second position. FIG. **9**B is a cross-sectional view of the reference side spindle flange **21** sectioned at the plane including the shaft line of the first roll sheet R**1** (shaft line of the spindle **19**), and FIG. **9**A is a cross-sectional view at the A-A line in FIG. **9**B. In the attitude at the second position, the first pressing portion **161** is pressing the flat portion **152** of the first contact portion **136**. The first position and the second position can be switched by operating the lever portion **163** in the rotating direction around the shaft line of the spindle **19**.

[0076] To attach the reference side spindle flange **21** to the first roll sheet R**1**, the rotating cam **160** is switched to the first position, as illustrated in FIGS. **8**A and **8**B. The reference side flange core **132** is configured so as to not interfere with the first shaft hole **41** of the first roll sheet R**1**. The interference portion **153** of the first contact portion **136** is also configured so as to not interfere at all or to interfere very little with the first shaft hole **41** of the first roll sheet R**1**. The second pressing portion **162** is also formed so as to not interfere with the first shaft hole **41** of the first roll sheet R**1**. Therefore, when the reference side spindle flange **21** is attached to the first roll sheet R**1**, friction between the interference portion **153** and the inner periphery of the first shaft hole **41** of the first roll sheet R**1** can be decreased, and operability is improved.

[0077] After the reference side spindle flange **21** is inserted into the first roll sheet R**1**, the rotating cam **160** is switched to the second position, as illustrated in FIGS. **9**A and **9**B. At the second position, the first pressing portion **161** presses the flat portion **152** of the first contact portion **136**. Then the interference portion **153** contacts the first shaft hole **41** of the first roll sheet R**1**, and the reference side spindle flange **21** is fixed to the first shaft hole **41** of the first roll sheet R**1**. The inner diameter of the first shaft hole **41** of the first roll sheet R**1** varies (large or small inner diameters), depending on the type of sheet, the manufacturing lot, and the like. However, the first contact portion **136** is an elastic body and can be elastically deformed, whereby variations of the inner diameter of the first shaft hole **41** of the first roll sheet R**1** can be absorbed, and the reference side spindle flange **21** can be fixed to the first roll sheet R**1**.
[0078] The non-reference side spindle flange **22** also has a configuration similar to the reference side spindle flange **21**, and the non-reference side spindle flange **22** is also fixed to the first roll sheet R**1**. By the above-mentioned procedure, the first roll sheet R**1**, having the first shaft hole **41** of which inner diameter is 2 inches, can be attached to the roll support member **2**.
[0079] Next, attachment of a second roll sheet R**2**, having a second shaft hole **42** of which inner diameter is 3 inches (second diameter), will be described. As illustrated in FIG. **10**, an attachment member **168** includes a second friction member **32** and an attachment installation portion **167**. The second friction member **32** is constituted of an attachment core **166** and 2 (a pair of) second contact portions **137**, which are disposed symmetrically with respect to the shaft line of the spindle **19**.
[0080] FIG. **11** is an external view of a state where the attachment member **168** is attached to the reference side spindle flange **21**. The attachment installation portion **167** is inserted into an attachment installation hole portion **164**, and the attachment member **168** is fixed to the reference side spindle flange **21** by a snap-fit type, screw fastening type, or the like.
[0081] FIG. **12** is an external view of the second contact portion **137**. The second contact portion **137** includes a fixing portion **154** which is fixed to the attachment member **168**. The second contact portion **137** also includes a flat portion **156**, to which the above mentioned second pressing portion **162** is pressed. The second contact portion **137** also includes an interference portion **157** which contacts with the inner periphery of the hollow shaft **40** of the second roll sheet R**2**. Further, the rotating cam **160** is disposed inside the reference side spindle flange **21**, as mentioned above. Just like the case of attachment of the 2-inch standard first roll sheet R**1**, the attitude at the first position illustrated in FIG. **7**A and the attitude at the second position illustrated in FIG. **7**B can be switched.
[0082] FIG. **13**B is a cross-sectional view of the reference side spindle flange **21** sectioned at a plane including the shaft line of the second roll sheet R**2** (shaft line of the spindle **19**), and FIG. **13**A is a cross-sectional view at the B-B line in FIG. **13**B. FIGS. **13**A and **13**B indicate the attitude of the second contact portion **137** and the rotating cam **160** at the first position. In the attitude at the first position, the second pressing portion **162** and the flat portion **156** of the second contact portion **137** do not contact.
[0083] FIGS. **14**A and **14**B indicate the attitude of the second contact portion **137** and the rotating cam **160** at the second position. FIG. **14**B is a cross-sectional view of the reference side spindle flange **21** sectioned at the plane including the shaft line of the second roll sheet R**2** (shaft line of the spindle **19**), and FIG. **14**B is a cross-sectional view at the B-B line in FIG. **14**B. In the attitude at the second position, the second pressing portion **162** presses the flat portion **156** of the second contact portion **137** through an opening portion **165** disposed on the reference side flange core **132**. The first position and the second position can be switched by operating the lever portion **163** in the rotating direction around the shaft line of the spindle **19**.
[0084] To attach the reference side spindle flange **21** to the second roll sheet R**2**, the rotating cam **160** is switched to the first position, as illustrated in FIGS. **13**A and **13**B. The attachment core **166** is configured so as to not interfere with the second shaft hole **42** of the second roll sheet R**2**. The interference portion **157** of the second contact portion **137** is also configured so as to not interfere at all or to interfere very little with the second shaft hole **42** of the second roll sheet R**2**. Therefore,

when the reference side spindle flange **21** is attached to the second roll sheet R**2**, friction between the interference portion **157** and the inner periphery of the second shaft hole **42** of the second roll sheet R**2**, can be decreased, and operability is improved.

[0085] After the reference side spindle flange **21** is inserted into the second roll sheet R**2**, the rotating cam **160** is switched to the second position, as illustrated in FIGS. **14**A and **14**B. At the second position, the second pressing portion **162** presses the flat portion **156** of the second contact portion **137**. Then the interference portion **157** contacts the second shaft hole **42** of the second roll sheet R**2**, and the reference side spindle flange **21** is fixed to the second shaft hole **42** of the second roll sheet R**2**.

[0086] The inner diameter of the second shaft hole **42** of the second roll sheet R**2** varies (large or small inner diameters), depending on the type of sheet, the manufacturing lot, and the like. However, the second contact portion **137** is an elastic body and can be elastically deformed, whereby variations of the inner diameter of the second shaft hole **42** of the second roll sheet R**2** can be absorbed, and the reference side spindle flange **21** can be fixed to the second roll sheet R**2**.

[0087] The non-reference side spindle flange **22** also has a configuration similar to the reference side spindle flange **21**, and the non-reference side spindle flange **22** is also fixed to the second roll sheet R**2**. By the above-mentioned procedure, the second roll sheet R**2**, having the second shaft hole **42** of which inner diameter is 3 inches, can be attached to the roll support member **2**.

[0088] As described above, in Embodiment 1, the first roll sheet R**1**, having the first shaft hole **41** of which inner diameter is the first inner diameter (e.g. 2 inches), and the second roll sheet R**2**, having the second shaft hole **42** of which inner diameter is the second inner diameter (e.g. 3 inches) which is larger than the first inner diameter, can be attached. The roll set portion **200** (holding device) includes the spindle **19**, which is a shaft member inserted into the shaft hole of the roll sheet, and the spindle holders **24** which are bearings to support both ends of the spindle **19**.

[0089] The reference side spindle flange **21** is the first fixing member having the reference side flange core **132**, which is the first core that can be inserted into the first shaft hole **41** of the first roll sheet R**1**. The first contact portion **136**, made of elastic, is disposed on the reference side spindle flange **21**, and can move to the first pressing position indicated in FIGS. **9**A and **9**B and the first non-pressing position indicated in FIGS. **8**A and **8**B. In the first pressing position, the first contact portion **136** contacts the inner peripheral surface of the first shaft hole **41** of the first roll sheet R**1**, and radially presses the inner peripheral surface. The first non-pressing position is a position located on the radially inward side of the first pressing position. The rotating cam **160** is a cam which moves the first contact portion **136** between the first pressing position and the first non-pressing position in a state where the attachment member **168** is not attached to the reference side spindle flange **21**.

[0090] The attachment member **168** is the second fixing member which can be attached to/detached from the reference side spindle flange **21**. The attachment member **168** includes the attachment core **166**, which is the second core that can be inserted into the second shaft hole **42** of the second roll sheet R**2**. The second contact portion **137**, made of elastic, is disposed on the attachment member **168**, and can move to the second pressing position indicated in FIGS. **14**A and **14**B and the second non-pressing position indicated in FIGS. **13**A and **13**B. At the second pressing position, the second contact portion **137** contacts the inner peripheral surface of the second shaft hole **42** of the second roll sheet R**2**, and presses the inner peripheral surface radially outward. The second non-pressing position is a position located on the radially inward side of the second pressing position. The rotating cam **160** moves the second contact portion **137** between the second pressing position and the second non-pressing position in a state where the attachment member **168** is attached to the reference side spindle flange **21**.

[0091] The rotating cam **160** can move between the first position indicated in FIGS. **7**A, **8**A, **8**B, **13**A and **13**B and the second position indicated in FIGS. **7**B, **9**A, **9**B, **14**A and **14**B. The rotating cam **160** includes the first pressing portion **161** and the second pressing portion **162**. In the state where the attachment member **168** is not attached to the reference side spindle flange **21**, the first pressing portion **161** does not contact the first contact portion **136** at the first position. The first pressing portion **161** contacts the first contact portion **136** at the second position, and presses the first contact portion **136** radially outward. In the state where the attachment member **168** is attached to the reference side spindle flange **21**, the second pressing portion **162** does not contact the second contact portion **137** at the first position. The second pressing portion **162** contacts the second contact portion **137** at the second position, and presses the second contact portion **137** radially outward.

[0092] The rotating cam **160** includes the cylindrical portion **169**, which is coaxial with the spindle **19** and to which the spindle **19** is inserted, and can move between the first position and the second position by the cylindrical portion **169** rotating around the spindle **19**. The first contact portion **136** and the second contact portion **137** are disposed at different positions in the axial direction of the spindle **19**, and at some positions (phases) in the circumferential direction of the spindle **19**. In Embodiment 1, the second contact portion **137** is disposed at a position that is closer to the center of the spindle **19** than the first contact portion **136** in the axial direction. In other words, the second contact portion **137** is disposed closer to the inlet of the shaft hole of the roll sheet (inlet on the side where the reference side spindle flange **21** is fixed) than the first contact portion **136** in the axial direction. The first pressing portion **161** and the second pressing portion **162** are disposed at positions corresponding to the first contact portion **136** and the second contact portion **137** respectively on the cylindrical portion **169** in the axial direction. The first contact portion **136** and the second contact portion **137** are metal leaf springs respectively. The lever portion **163** is an operation member with which the user can perform operation to move the rotating cam **160**.

[0093] In the case of setting the first roll sheet R**1** in the roll set portion **200**, the reference side flange core **132** is inserted into the first shaft hole **41** in a state where the attachment member **168** is not attached to the reference side spindle flange **21** and the first contact portion **136** is at the first non-pressing position. Then the first contact portion **136** is moved to the first pressing position by the rotating cam **160**, and the reference side spindle flange **21** is fixed to the first shaft hole **41**, whereby the first roll sheet R**1** is fixed to the spindle **19**. In the state where the first roll sheet R**1** is set in the roll set portion **200**, the print head **8** of the recording device **100** performs recording on the sheet **1** (first sheet) supplied from the first roll sheet R**1**.

[0094] In the case of setting the second roll sheet R**2** in the roll set portion **200**, the attachment core **166** is inserted into the second shaft hole **42** in a state where the attachment member **168** is attached to the reference side spindle flange **21**, and the second contact portion **137** is at the second non-pressing position. Then the second contact portion **137** is moved to the second pressing position by the rotating cam **160**, and the attachment member **168** is fixed to the second shaft hole **42**, whereby the second roll sheet R**2** is fixed to the spindle **19**. In the state where the second roll sheet R**2** is set in the roll set portion **200**, the print head **8** of the recording device **100** performs recording on the sheet **1** (second sheet) supplied from the second roll sheet R**2**.

[0095] The first contact portion **136** moves to the first non-pressing position by moving the rotating cam **160** to the first position in the state where the attachment member **168** is not attached to the reference side spindle flange **21**. The first contact portion **136** moves to the first pressing position by moving the rotating cam **160** to the second position.

[0096] The second contact portion **137** moves to the second non-pressing position by moving the rotating cam **160** to the first position in the state where the attachment member **168** is attached to the reference side spindle flange **21**. The second contact portion **137** moves to the second pressing position by moving the rotating cam **160** to the second position.

[0097] As described above, in the holding device which can use a plurality of rolled recording media having different inner diameters (e.g. 2 inches, 3 inches), Embodiment 1 allows to decrease the force required to insert or extract the sheet to attached to or detach from the holding device. Thereby operability in the operation to attach or detach the sheet to/from the holding device can be improved.

Embodiment 2

[0098] Now the configuration of Embodiment 2, which allows to attach roll sheets having different inner diameters of shaft hole of the shafts to the roll set portion **200** (holing device), will be described with reference to FIGS. **15** to **25**.

[0099] First, attachment of a first roll sheet R**1** having a first shaft hole **41** of which inner diameter is 2 inches (first inner diameter) will be described.

[0100] FIG. **15** is an external view of a spindle flange. The reference side spindle flange **21** includes the flat reference side flange surface **131** and the attachment installation hole portions **164**. The first friction member **31** is constituted of the reference side flange core **132** which protrudes from the reference side flange surface **131**, and a plurality of first contact portions **136** which are disposed symmetrically with respect to the shaft line of the spindle **19**.

[0101] FIG. **16** is an external view of the first contact portion **136**. The first contact portion **136** includes the fixing portion **150** which is fixed to the reference side spindle flange **21**. The first contact portion **136** also includes the flat portion **152** to which a later mentioned pressing portion **171** is pressed. The first contact portion **136** also includes an interference portion **153**, which contacts with the inner periphery of the first shaft hole **41** of the first roll sheet R**1**.

[0102] A linearly moving cam **170** is disposed inside the reference side spindle flange **21**. FIG. **17** is an external view of the linearly moving cam **170**. The linearly moving cam **170** includes: a cylindrical portion **179** which is coaxial with the spindle **19**; a flat portion **172** to which a later mentioned contact portion **176** and contact portion **177** contact; and a ring-shaped flat portion **173** to which a later mentioned compression spring **178** contacts. The end portion of the cylindrical portion **179** functions as a pressing portion **171**. The linearly moving cam **170** is supported inside the reference side spindle flange **21**, so as to be linearly movable.

[0103] FIG. **18** is an external view of a grip portion **174**. The grip portion **174** includes a bearing portion **175** which is connected with the reference side spindle flange **21** and rotates, the contact portion **176** and the contact portion **177**. The contact portion **176** and the contact portion **177** are parallel with the rotation center shaft of the bearing portion **175**, and the contact portion **176** is disposed closer to the rotation center shaft of the bearing portion **175** than the contact portion **177**.

[0104] FIG. **19** is a cross-sectional view of the reference side spindle flange **21** sectioned at a plane including the shaft line of the first roll sheet R**1** (shaft line of the spindle **19**), and indicates the attitude of the first contact portion **136**, the linearly moving cam **170**, and the grip portion **174** at the first position. At the first position, the pressing portion **171** does not contact the flat portion **152** of the first contact portion **136**.

[0105] FIG. **20** indicates the attitude of the first contact portion **136**, the linearly moving cam **170** and the grip portion **174** at the second position. At the second position, the pressing portion **171** presses the flat portion **152** of the first contact portion **136**.

[0106] There is a cylindrical-shaped gap between the first contact portion **136** and the linearly moving cam **170**, and the compression spring **178** is disposed in this gap. When force is applied from the compression spring **178** in the direction of releasing the compression, the contact portion **176** of the grip portion **174** contacts the flat portion **172** of the linearly moving cam **170** at the first position, and the contact portion **177** of the grip portion **174** contacts the flat portion **172** of the linearly moving cam **170** at the second position.

[0107] By rotating the grip portion **174** in the direction of C**3** in FIG. **19**, the contact position of the grip portion **174** is switched from the contact portion **176** to the contact portion **177**. Then the linearly moving cam **170** linearly moves to the left in FIG. **19**, and pushes up the first contact portion **136** in the radius direction of the first roll sheet R**1**, whereby the first position is switched to the second position.

[0108] By rotating the grip portion **174** in the direction of C**4** in FIG. **20**, the contact position of the grip portion **174** is switched from the contact portion **177** to the contact portion **176**, then the linearly moving cam **170** linearly moves to the right in FIG. **20**, whereby the second position is switched to the first position.

[0109] To attach the reference side spindle flange **21** to the first roll sheet R**1**, the linearly moving cam **170** is switched to the first position, as illustrated in FIG. **19**. The reference side flange core **132** is configured so as to not interfere with the first shaft hole **41** of the first roll sheet R**1**. The interference portion **153** of the first contact portion **136** is also configured so as to not interfere at all or to interfere very little with the first shaft hole **41** of the first roll sheet R**1**. Therefore, when the reference side spindle flange **21** is attached to the first roll sheet R**1**, friction between the interference portion **153** and the inner periphery of the first shaft hole **41** of the first roll sheet R**1** can be decreased, and operability is improved.

[0110] After the reference side spindle flange **21** is inserted into the first roll sheet R**1**, the linearly moving cam **170** is switched to the second position, as illustrated in FIG. **20**. As mentioned above, at the second position, the pressing portion **171** presses the flat portion **152** of the first contact portion **136**. Then the interference portion **153** contacts the first shaft hole **41** of the first roll sheet R**1**, and the reference side spindle flange **21** is fixed to the first shaft hole **41** of the first roll sheet R**1**.

[0111] The inner diameter of the first shaft hole **41** of the first roll sheet R**1** varies (large or small inner diameters), depending on the type of sheet, the manufacturing lot, and the like. However, the first contact portion **136** is an elastic body and can be elastically deformed, whereby variations of the inner diameter of the first shaft hole **41** of the first roll sheet R**1** can be absorbed, and the reference side spindle flange **21** can be fixed to the first roll sheet R**1**.

[0112] The non-reference side spindle flange **22** also has a configuration similar to the reference side spindle flange **21**, and the non-reference side spindle flange **22** is also fixed to the first roll sheet R**1**. By the above-mentioned procedure, the first roll sheet R**1**, having the first shaft hole **41** of which inner diameter is 2 inches, can be attached to the roll support member **2**.

[0113] Next, attachment of the second roll sheet R**2**, having a second shaft hole **42** of which inner diameter is 3 inches (second diameter), will be described. As illustrated in FIG. **21**, an attachment member **168** includes the second friction member **32** and the attachment installation portion **167**. The second friction member **32** is constituted of an attachment core **166**, and a plurality of second contact portions **137**, which are disposed symmetrically with respect to the shaft line of the spindle **19**.

[0114] FIG. **22** is an external view of a state where the attachment member **168** is attached to the reference side spindle flange **21**. The attachment installation portion **167** is inserted into the attachment installation hole portion **164**, and the attachment member **168** is fixed to the reference side spindle flange **21** by a snap-fit type, screw fastening type, or the like. The second contact portion **137** is disposed in the same rotation phase as the opening portion **165** indicated in FIG. **15**, with respect to the shaft line of the spindle **19**.

[0115] FIG. **23** is an external view of the second contact portion **137**. The second contact portion **137** includes the fixing portion **154** which is fixed to the attachment member **168**. The second contact portion **137** also includes the flat portion **156**, to which the above mentioned second pressing portion **162** is pressed. The second contact portion **137** also includes the interference portion **157** which contacts with the inner periphery of the second shaft hole **42** of the second roll sheet R**2**. Further, the linearly moving cam **170** is disposed inside the reference side spindle flange **21**, and just like the case of the attachment of the 2 inch standard first roll sheet R**1**, the attitude at the first position illustrated in FIG. **19** and the attitude at the second position illustrated in FIG. **20** can be switched.

[0116] FIGS. **24** and **25** are cross-sectional views of the reference side spindle flange **21** sectioned at a plane including the shaft line of the second roll sheet R**2** (shaft line of the spindle **19**). FIG. **24** indicates the attitude of the second contact portion **137** and the linearly moving cam **170** at the first position. Ay the first position, the pressing portion **171** and the flat portion **156** of the second contact portion **137** do not contact. FIG. **25** indicates the attitude of the second contact portion **137** and the linearly moving cam **170** at the second position. At the second position, the pressing portion **171** presses the flat portion **156** of the second contact portion **137** through the opening portion **165** disposed in the reference side flange core **132**. Just like the case of attaching the 2-inch standard first roll sheet R**1**, the first position and the second position can be switched by operating the grip portion **174** in the C**3** direction in FIG. **24** and the C**4** direction in FIG. **25**.

[0117] To attach the reference side spindle flange **21** to the second roll sheet R**2**, the linearly moving cam **170** is switched to the first position, as illustrate din FIG. **24**. The attachment core **166** is configured so as to not interfere with the second shaft hole **42** of the second roll sheet R**2**. The interference portion **157** of the second contact portion **137** is also configured so as to not interfere at all or to interfere very little with the second shaft hole **42** of the second roll sheet R**2**. Therefore, when the reference side spindle flange **21** is attached to the second roll sheet R**2**, friction between the interference portion **157** and the inner periphery of the second shaft hole **42** of the second roll sheet R**2** can be decreased, and operability is improved.

[0118] After the reference side spindle flange **21** is inserted into the second roll sheet R**2**, the linearly moving cam **170** is switched to the second position, as illustrated in FIG. **25**. At the second position, the pressing portion **171** presses the flat portion **156** of the second contact portion **137**. Then the interference portion **157** contacts the second shaft hole **42** of the second roll sheet R**2**, and the reference side spindle flange **21** is fixed to the second shaft hole **42** of the second roll sheet R**2**.

[0119] The inner diameter of the second shaft hole **42** of the second roll sheet R**2** varies (large or small inner diameters), depending on the type of sheet, the manufacturing lot, and the like. However, the second contact portion **137** is an elastic body and can be elastically deformed, whereby variations of the inner diameter of the second shaft hole **42** of the second roll sheet R**2** can be absorbed, and the reference side spindle flange **21** can be fixed to the second roll sheet R**2**.

[0120] The non-reference side spindle flange **22** also has a configuration similar to the reference side spindle flange **21**, and the non-reference side spindle flange **22** is also fixed to the second roll sheet R**2**. By the above-mentioned procedure, the second roll sheet R**2**, having the second shaft hole **42** of which inner diameter is 3 inches, can be attached to the roll support member **2**.

[0121] As described above, in Embodiment 2, the linearly moving cam **170** can move between the first position indicated in FIGS. **19** and **24**, and the second position indicated in FIGS. **20** and **25**. The linearly moving cam **170** includes the pressing portion **171**. In the state where the attachment member **168** is not attached to the reference side spindle flange **21**, the pressing portion **171** does not contact the first contact portion **136** at the first position. The pressing portion **171** contacts the first contact portion **136** at the second position, and presses the first contact portion **136** radially outward. In the state where the attachment member **168** is attached to the reference side spindle flange **21**, the pressing portion **171** does not contact the second contact portion **137** at the first position. The pressing portion **171** contacts the second contact portion **137** in the second position, and presses the second contact portion **137** radially outward.

[0122] The pressing portion **171** of the linearly moving cam **170** is an end portion of the cylindrical portion **179**, which is coaxial with the spindle **19** and to which the spindle **19** is inserted, and the linearly moving cam **170** can move between the first position and the second position by the cylindrical portion moving in parallel with the axial direction of the spindle **19**. The first contact portion **136** and the second contact portion **137** are disposed at different positions (phases) in the circumferential direction of the spindle **19**. The first contact portion **136** and the second contact portion **137** are metal leaf springs respectively. The grip portion **174** is an operation member with which the user can perform operation to move the linearly moving cam **170**.

[0123] In the case of setting the first roll sheet R**1** in the roll set portion **200**, the reference side flange core **132** is inserted into the first shaft hole **41** in a state where the attachment member **168** is not attached to the reference side spindle flange **21**, and the first contact portion **136** is at the first non-pressing position. Then the first contact portion **136** is moved to the first pressing position by the linearly moving cam **170**, and the reference side spindle flange **21** is fixed to the first shaft hole **41**, whereby the first roll sheet R**1** is fixed to the spindle **19**.

[0124] In the case of setting the second roll sheet R**2** in the roll set portion **200**, the attachment core **166** is inserted into the second shaft hole **42** in a state where the attachment member **168** is attached to the reference side spindle flange **21**, and the second contact portion **137** is at the second non-pressing position. Then the second contact portion **137** is moved to the second pressing position by the linearly moving cam **170**, and the attachment member **168** is fixed to the second shaft hole **42**, whereby the second roll sheet R**2** is fixed to the spindle **19**.

[0125] The first contact portion **136** moves to the first non-pressing position by moving the linearly moving cam **170** to the first position in the state where the attachment member **168** is not attached to the reference side spindle flange **21**. The first contact portion **136** moves to the first pressing position by moving the linearly moving cam **170** to the second position.

[0126] The second contact portion **137** moves to the second non-pressing position by moving the linearly moving cam **170** to the first position in a state where the attachment member **168** is attached to the reference side spindle flange **21**. The second contact portion **137** moves to the second pressing position by moving the linearly moving cam **170** to the second position.

[0127] As described above, in the holding device which can use a plurality of rolled recording media having different inner diameters (e.g. 2 inches, 3 inches), Embodiment 2 allows to decrease the force required to insert or extract the sheet to attach to or detach from the holding device. Thereby operability in the operation to attach or detach the sheet to/from the holding device can be improved.

[0128] The first contact portion **136** and the second contact portion **137** are disposed at different rotational phases with respect to the shaft line of the spindle **19**. By this disposition, both the first contact portion **136** and the second contact portion **137** can be disposed in positions distant from the reference side flange surface **131**. Now a case of creating a small gap between the reference side flange surface **131** and the end portions of the first roll sheet R**1** and the second roll sheet R**2** in the width direction is considered. In this case, if both the first contact portion **136** and the second contact portion **137** are at positions distant from the reference side flange surface **131**, the reference side flange surface **131** and the end portions of the first roll sheet R**1** and the second roll sheet R**2** in the width direction contact at an inclination smaller than the case of being at positions close to the reference side flange surface **131**. Therefore if the first contact portion **136** and the second contact portion **137** are disposed in the positions more distant from the reference side flange surface **131**, it can be prevented that the reference side spindle flange **21** inclines from the shaft line of the first shaft hole **41** and the second shaft hole **42** of the first roll sheet R**1** and the second roll sheet R**2**. Based on this relationship, both the first contact portion **136** and the second contact portion **137** are disposed in positions distant from the reference side flange surface **131**, whereby inclination of the reference side spindle flange **21** from both the first roll sheet R**1** and the second roll sheet R**2** can be prevented. This is the same for the non-reference side spindle flange **22**.

Embodiment 3

[0129] In Embodiment 3, a lock structure to fix the spindle **19** and the non-reference side spindle flange **22** will be described. Further, a lock structure that allows to perform the above fixing operation together with the fixing operation of the non-reference side spindle flange **22** and the roll sheet R, described in Embodiment 1, will be described. FIGS. **26** to **31** are diagrams for describing the non-reference side spindle flange **22** according to Embodiment 3. In the following, differences from Embodiment 1 will be mainly described, and description on the configurations the same as Embodiment 1 will be omitted.

[0130] First the structure to lock the non-reference side spindle flange **22** and the spindle **19** will be described with reference to FIGS. **26** to **30**.

[0131] FIG. **26** is a diagram indicating a spindle lock structure installed in the non-reference side spindle flange **22**. In addition to the lock structure of the first shaft hole **41** of the first roll sheet R**1** and the reference side spindle flange **21** described in Embodiment 1, Embodiment 3 includes the lock structure of the spindle **19** and the non-reference side spindle flange **22**. As illustrated in FIG. **26**, the non-reference side spindle flange **22** includes a spindle lock flange **231** and a spindle lock lever **232**. By rotating the spindle lock lever **232** clockwise, a later mentioned spindle interference portion **236** is contracted, whereby the spindle **19** is fixed. This lock structure is a structure which is disposed only io the non-reference side spindle flange **22**, and is not disposed on the reference side spindle flange **21**, which is fixed to the spindle **19** without separation.

[0132] FIG. **27** is an external view of the spindle lock flange **231**. The spindle lock flange **231** includes a flange cavity portion **233**. In the flange cavity portion **233**, a member used for transferring driving to perform locking is housed. The flange cavity portion **233** includes a leaf spring for locking **234**, and a spring for locking **235**, whereby the force at driving the lever is transferred and a spindle interference portion **236** of the spindle lock flange **231** is contracted, so as to implement the spindle lock. The leaf spring for locking **234** is fixed to the spindle lock lever **232** coaxially through a fixing shaft **237**.

[0133] FIG. **28** is an external view of the spindle lock lever **232**. The spindle lock lever **232** is fixed to the leaf spring for locking **234** through a fixing shaft installation hole portion **238**. The base of the spindle lock lever **232** includes a cam flat portion **252** and a cam curved face portion **253**. When the spindle lock lever **232** is rotated around the fixing shaft **237**, the cam curved face portion **253** presses the flat portion **251** of the spindle lock flange **231** and creates the locked state.

[0134] FIG. **29** is an external view of the non-reference side spindle flange **22** in a state where the spindle lock is not performed. In the unlocked state, the base of the spindle lock lever **232** is in a state where the cam flat portion **252** faces a flat portion **251** of the spindle lock flange **231**, and the two members do not interfere with each other.

[0135] FIG. **30** is an external view of the non-reference side spindle flange **22** in a state where the spindle lock is performed. To perform the spindle lock, the spindle lock lever **232** is rotated clockwise in FIG. **30** around the fixing shaft **237** (rotation shaft), so that the cam curved face portion **253** is pressing the flat portion **251** of the spindle lock flange **231** at the base of the spindle lock lever **232**.

[0136] To insert the non-reference side spindle flange **22** to the spindle **19**, the spindle lock lever **232** is switched to the unlocked state, as illustrated in FIG. **29**, and the first roll sheet R**1** is attached in this state. After attaching the first roll sheet R**1**, the non-reference side spindle flange **22** is inserted into the first roll sheet R**1**, and the spindle lock lever **232** is rotated, as indicated in FIG. **30**, to lock the state. This lock structure can function in the same manner even when a 3-inch dedicated attachment member **168** is attached.

[0137] After the non-reference side spindle flange **22** is inserted into the first shaft hole **41** of the first roll sheet R**1**, the lever portion **163** can be rotated for sheet tube lock, which fixes the non-reference side spindle flange **22** to the first shaft hole **41**. Further, the spindle lock lever **232** can be rotated for spindle lock, which fixes the non-reference side spindle flange **22** to the spindle **19**. These rotating operations are performed independently from each other. Therefore, to set the first roll sheet R**1** in the printer, two lock operations (operation to perform the sheet tube lock by rotating the lever portion **163**, and operation to perform the spindle lock by rotating the spindle lock level **232**) are required. To prevent this, a configuration to perform both the sheet tube lock and the spindle lock simultaneously by performing only one of the sheet tube lock and the spindle lock will be described with reference to FIGS. **31** and **32**.

[0138] The lever portion **163** and the spindle lock lever **232** are connected with an intermediate link **256** and an intermediate link **257**, and the lever portion **163** is rotated from a first position to a second position in this state. At this time, rotational driving is also transferred to the spindle lock lever **232**, whereby the sheet tube lock and the spindle lock can be performed simultaneously.

[0139] FIG. **31** is a diagram indicating the non-reference side spindle flange **22** in a case where both the sheet tube and the spindle are in the unlocked state. FIG. **32** is a diagram indicating the

non-reference side spindle flange **22** in a case where both the sheet tube and the spindle are in the locked state. If the lever portion **163** is rotated clockwise in FIG. **31**, the rotational driving is transferred to the spindle lock lever **232** via the intermediate link **256** and the intermediate link **257**, whereby the spindle lock lever **232** is also rotated clockwise. In this example, the linking is the mechanism to transfer the driving acting on the lever portion **163** to the spindle lock lever **232**, but the transfer member may be constituted of a gear, a cam or the like. The sheet tube lock is not limited to the operation to rotate the lever portion **163**, but may be a configuration where such a lock mechanism as the grip portion **174** interlocks with the spindle lock lever **232**, for example. In this way, by implementing a configuration in which two locks are established simultaneously with one operation, the user operation to set the first roll sheet R**1** is simplified. The above description is also applicable to the setting of the second roll sheet R**2**.
[0140] As described above, in Embodiment 3, the non-reference side spindle flange **22** (first fixing member), to fix the roll sheet R to the spindle **19**, has a through hole **293** to which the spindle **19** is inserted. The spindle lock flange **231**, the spindle lock lever **232** and the spindle interference portion **236** constitutes a fastening portion which radially fastens the spindle **19** inserted into the through hole **293** so that the fastening portion is able to fix the non-reference side spindle flange **22** to the spindle **19**. In a state where the attachment member **168** is not attached to the reference side spindle flange **21**, the lever portion **163** can be operated to move the rotating cam **160** from the first non-pressing position to the first pressing position (a first operation). Further, in a state where the attachment member **168** is attached to the reference side spindle flange **21**, the lever portion **163** can be operated to move the rotating cam **160** from the second non-pressing position to the second pressing position (a second operation). The intermediate links **256** and **257** are mechanisms to interlock the first and second operations with an action of fastening the spindle **19** by the fastening portion.
[0141] According to the present disclosure, in the recording device having the holding device which can support roll sheets of which inner diameters of the respective shaft holes are different, operability of the attachment/detachment of the roll sheet to/from the holding device can be improved.
[0142] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
[0143] This application claims the benefit of Japanese Patent Application No. 2024-033235, filed on Mar. 5, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A recording device, comprising: a holding device configured to hold a roll sheet that is a sheet rolled around a shaft having a shaft hole; and a recording unit configured to perform recording on the sheet supplied from the roll sheet held by the holding device, wherein the holding device further comprising: a first fixing member including a first core configured to be inserted into a first shaft hole of a first roll sheet, the first shaft hole having a first inner diameter; a first contact portion formed of an elastic body disposed in the first fixing member, the first contact portion being configured to move between a first pressing position where the first contact portion contacts an inner peripheral surface of the first shaft hole and presses the inner peripheral surface radially outward and a first non-pressing position located on a radially inward side of the first pressing position; a cam configured to move the first contact portion between the first pressing position and the first non-pressing position; a second fixing member configured to be attached to or detached from the first fixing member, the second fixing member including a second core configured to be inserted into a second shaft hole of a second roll sheet, the second shaft hole having a second inner diameter larger than the first inner diameter; and a second contact portion formed of an elastic body disposed in the second fixing member, the second contact portion being configured to move between a second pressing position where the second contact portion contacts an inner peripheral surface of the second shaft hole and presses the second shaft hole radially outward and a second non-pressing position located on a radially inward side of the second pressing position, wherein in a state where the second fixing member is attached to the first fixing member, the cam moves the second contact portion between the second pressing position and the second non-pressing position.

2. The recording device according to claim 1, wherein the recording unit is configured to perform recording on a first sheet supplied from the first roll sheet, into the first shaft hole of which the first core of the first fixing member is inserted.

3. The recording device according to claim 1, wherein the recording unit is configured to perform recording on a second sheet supplied from the second roll sheet, into the second shaft hole of which the second core of the second fixing member is inserted.

4. The recording device according to claim 1, wherein the recording unit performs recording on a sheet by ejecting ink.

5. The recording device according to claim 1, wherein the cam is configured to move between a first position and a second position, and the cam includes: a first pressing portion configured not to contact the first contact portion at the first position and to contact the first contact portion and press the first contact portion radially outward at the second position, in a state where the second fixing member is not attached to the first fixing member; and a second pressing portion configured not to contact the second contact portion at the first position and to contact the second contact portion and press the second contact portion radially outward at the second position, in a state where the second fixing member is attached to the first fixing member.

6. The recording device according to claim 5, wherein the recording device includes: a shaft member configured to be inserted into the shaft hole of the roll sheet; and bearings configured to support both ends of the shaft member, wherein the cam includes a cylindrical portion to which the shaft member is inserted and which is coaxial with the shaft member, and the cam is configured to move between the first position and the second position by the cylindrical portion rotating around the shaft member.

7. The recording device according to claim 6, wherein the first contact portion and the second contact portion are disposed at different positions in an axial direction of the shaft member, and the first pressing portion and the second pressing portion are disposed at positions corresponding to the first contact portion and the second contact portion respectively in the axial direction of the cylindrical portion.

8. The recording device according to claim 7, wherein the second contact portion is disposed at a position closer to the center of the shaft member than the first contact portion in the axial direction.

9. The recording device according to claim 6, wherein the first contact portion and the second contact portion are disposed at the same position in a circumferential direction of the shaft member.

10. The recording device according to claim 1, wherein the cam is configured to move between a first position and a second position, the cam includes a pressing portion, the pressing portion is configured not to contact the first contact portion at the first position and to contact the first contact portion and press the first contact portion radially outward at the second position, in a state where the second fixing member is not attached to the first fixing member, and the pressing portion is configured not to contact the second contact portion in the first position and to contact the second contact portion and press the second contact portion radially outward at the second position, in a state where the second fixing member is attached to the first fixing member.

11. The recording device according to claim 10, wherein the recording device includes: a shaft member configured to be inserted into the shaft hole of the roll sheet; and bearings configured to support both ends of the shaft member, wherein the cam includes a cylindrical portion to which the shaft member is inserted and which is coaxial with the shaft member, and the cam is configured to move between the first position and the second position by the cylindrical portion moving parallel with the axial direction of the shaft member.

12. The recording device according to claim 11, wherein the first contact portion and the second contact portion are disposed at different positions in a circumferential direction of the shaft member, and the pressing portion is an end portion of the cylindrical portion in the axial direction.

13. The recording device according to claim 1, wherein the first contact portion and the second contact portion are metal leaf springs respectively.

14. The recording device according to claim 1, further comprising an operation member with which a user performs operation to move the cam.

15. The recording device according to claim 14, wherein the recording device includes: a shaft member configured to be inserted into the shaft hole of the roll sheet; and bearings configured to support both ends of the shaft member, wherein the first fixing member includes: a through hole through which the shaft member is inserted; and a fastening portion configured to fix the first fixing member to the shaft member by radially fastening the shaft member inserted in the through hole, wherein the recording device includes a mechanism configured to interlock first and second operations of the operation member with an action of fastening the shaft member by the fastening portion, the first operation being an operation of the operation member for moving the cam from the first non-pressing position to the first pressing position in a state where the second fixing member is not attached to the first fixing member, and the second operation being an operation of the operation member for moving the cam from the second non-pressing position to the second pressing position in a state where the second fixing member is attached to the first fixing member.